Calculus 12 - Derivative Techniques - Practice Test v2

Rubric								
Outcome	0	1	2	3	4	5	6	7
Understanding and Solving								
Mathematical Communication								

St. George's School - Academic Integrity

Academic Dishonesty includes any conduct with the intent to gain an unfair advantage in connection with an academic assessment. Academic Dishonesty can occur in many ways. Common forms of Academic Dishonesty include, but are not limited to, cheating, falsification, plagiarism, and tampering. Cheating occurs when an individual undermines the integrity of an assessment (including homework and other assignments, reports, projects, quizzes, tests, exams, or other forms of performance evaluations).

Examples of cheating include:

- Copying any part of an assessment;
- Allowing others to copy any part of an assessment;
- Improperly giving or receiving assessment information;
- Using unauthorized resources for or during an assessment;
- Skipping classes to avoid an assessment

Name : _____

Teacher's name: _____

Level 1 to Level 2 Questions

1. Differentiate the following functions. No need to simplify.

   (a) $y = 3x^7 - 4x^5 + 10x - 6$

   (b) $f(x) = (x^2 - 3x)(x^2 + 4x + 1)$

   (c) $g(x) = \dfrac{x^{3/2} - x}{x^2 + 1}$

   (d) $y = \sqrt{4x^2 - 9x + 2}$

2. Use implicit differentiation to find $\dfrac{dy}{dx}$ for the following

$$2x^2 + 5y - y^3 = 8$$

Level 3 Questions

3. Find the equation of the tangent line to the curve $y = (3x + 1)^3$ at the point $(0, 1)$.

4. Show that the function $f(x) = x^3 - 6x^2 + 14x - 4$ has no horizontal tangent.

5. Using the table of values below, determine the following.

x	$f(x)$	$f'(x)$	$h(x)$	$h'(x)$
3	5	-4	5	3
5	-2	6	-3	1

   (a) Find the derivative of $y = f(x)h(x)$ at $x = 3$

   (b) Find the derivative of $y = f(h(x))$ at $x = 3$

Level 4 to Level 6 Questions

6. Compute $\dfrac{dy}{dx}$ for the following functions. No need to simplify.

   (a) $y = \sqrt{\dfrac{3x+1}{2x^2+5}}$

   (b) $y = \sqrt{3x+2} \cdot (x^3-1)^5$

7. Use implicit differentiation to find y' for the following relation.

$$\sqrt{x+1} + x^2 y = y^{3/2}$$

8. Determine where the following function is not differentiable. Explain using the conditions for differentiability.

$$f(x) = \begin{cases} x+1 & \text{if } x < 0 \\ x^2 + x + 1 & \text{if } 0 \leq x < 3 \\ 7x - 9 & \text{if } x \geq 3 \end{cases}$$

9. Find the second derivative of the function $y = \sqrt{x^2 - 5x + 4}$

Level 7 Questions

10. Solve for a and b such that the following piecewise function is **differentiable** everywhere.

$$f(x) = \begin{cases} ax^2 + 2x & \text{if } x < 1 \\ bx + 6 & \text{if } x \geq 1 \end{cases}$$

11. Given the curve $y = \dfrac{1}{x}$, give a simplified expression for the nth derivative of y.

hint: $n! = n \times (n-1) \times (n-2) \times ... \times 3 \times 2 \times 1$

ex. $5! = 5 \times 4 \times 3 \times 2 \times 1$